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Exercise: SQL Joins Exercise

Part A – Inner Join

Question 1

Select A.user_id,
user_name,
subscription_id,
start_date
From users AS A
Inner Join subscriptions AS B
ON A.user_id = B.user_id;

```
43
44  --Question 1: Show every user who has a subscription. Match users to subscriptions
45  Select A.user_id,
46  user_name,
47  subscription_id,
48  start_date
49  From joins.subs.users AS A
50  Inner Join joins1.subs1.subscriptions AS B
51  ON A.user_id = B.user_id;
52
```

Add parameter

Table

	user_id	user_name	subscription_id	start_date
1	1	Nomvula	503	2026-03-10
2	2	David	502	2026-02-01
3	3	Anele	505	2026-04-05
4	1	Nomvula	501	2026-01-15

Question 2

Select subscription_id,
user_id,
A.plan_id,
plan_name,
monthly_price
From subscriptions AS A
Inner Join plans AS B
ON A.plan_id = B.plan_id;

```sql
--Question 2:Show every subscription with its matching plan name and monthly price.

Select *
from joins1.subs1.subscriptions;

Select subscription_id,
user_id,
A.plan_id,
plan_name,
monthly_price
From joins1.subs1.subscriptions AS A
Inner Join joins2.subs2.plans AS B
```

Add parameter

Table +

	subscription_id	user_id	plan_id	plan_name	monthly_price
1	501	1	10	Basic	79
2	504	6	11	Standard	129
3	503	1	12	Premium	199
4	505	3	13	Family	249
5	502	2	11	Standard	129

Question 3

```
select session_id,
user_id,
A.show_id,
genre,
watch_minutes
from sessions as A
inner shows as B
on A.show_id = B.show_id;
```

```sql
--Question 3: Show every viewing session that has a matching show. Include the show title and genre.
select session_id,
user_id,
A.show_id,
genre,
watch_minutes
from joins4.subs4.sessions as A
inner join joins3.subs3.shows as B
on A.show_id = B.show_id;
```

Add parameter

Table

	session_id	user_id	show_id	genre	watch_minutes
1	904	7	701	Comedy	20
2	903	1	702	Drama	60
3	902	2	703	Documentary	30
4	901	1	701	Comedy	45

Question 4

Select A.user_id,
user_name,
country,
session_id,
show_id,
watch_minutes
From users AS A
Inner Join sessions AS B
ON A.user_id = B.user_id;

```sql
--Question 4:Show every viewing session with the user who watched it. Only show sessions with a matching user.

Select A.user_id,
user_name,
country,
session_id,
show_id,
watch_minutes
From joins.subs.users AS A
Inner Join joins4.subs4.sessions AS B
ON A.user_id = B.user_id;
```

Add parameter

Table

	user_id	user_name	country	session_id	show_id	watch_minutes
1	1	Nomvula	johannesburg	903	702	60
2	2	David	Cape Town	902	703	30
3	3	Anele	Durban	905	705	90
4	1	Nomvula	johannesburg	901	701	45

Question 5

Select A.user_id,
user_name,
country,
B.plan_id
plan_name,
monthly_price,
start_date
From users AS A
Inner subscriptions AS B
ON A.user_id = B.user_id
Inner plans AS C
ON B.plan_id = C.plan_id;

```
155
156  --Questions 5: Show users along with their subscriptions, the plan name, and the price. Use only users who have both a subscription and a valid plan.
157  Select A.user_id,
158  user_name,
159  country,
160  B.plan_id
161  plan_name,
162  monthly_price,
163  start_date
164  From joins.subs.users AS A
165  Inner Join joins1.subs1.subscriptions AS B
166  ON A.user_id = B.user_id
167  Inner Join joins2.subs2.plans AS C
168  ON B.plan_id = C.plan_id;
169
```

Add parameter

Table

	user_id	user_name	country	plan_name	monthly_price	start_date
1	1	Nomvula	johannesburg	12	199	2026-03-10
2	2	David	Cape Town	11	129	2026-02-01
3	3	Anele	Durban	13	249	2026-04-05
4	1	Nomvula	johannesburg	10	79	2026-01-15

PART B

Question 6

Select A.user_id,
user_name,
subscription_id,
start_date
From users AS A
Left Join subscriptions AS B
ON A.user_id = B.user_id

```sql
--Question 6: Show every user and any subscriptions they have. Users without subscriptions must still appear.
Select A.user_id,
user_name,
subscription_id,
start_date
From joins.subs.users AS A
Left Join joins1.subs1.subscriptions AS B
ON A.user_id = B.user_id
```

Add parameter

Table

	user_id	user_name	subscription_id	start_date
1	1	Nomvula	503	2026-03-10
2	2	David	502	2026-02-01
3	3	Anele	505	2026-04-05
4	4	Kabelo	null	null
5	5	Lerato	null	null
6	1	Nomvula	501	2026-01-15

Question 7

Select A.plan_id,
plan_name,
subscription_id,
user_id
From plans AS A
Left Join subscriptions AS B
ON A.plan_id = B.plan_id;

```sql
--Question 7: Show every plan and the subscriptions on it. Plans with no subscribers must still appear.
Select A.plan_id,
plan_name,
subscription_id,
user_id
From joins2.subs2.plans AS A
Left Join joins1.subs1.subscriptions AS B
ON A.plan_id = B.plan_id;
```

Add parameter

Table

	plan_id	plan_name	subscription_id	user_id
1	10	Basic	501	1
2	11	Standard	504	6
3	12	Premium	503	1
4	13	Family	505	3
5	14	Mobile	null	null
6	11	Standard	502	2

Question 8

```
select A.show_id,
show_title,
session_id,
watch_minutes
from shows as A
left join sessions as B
on A.show_id = B.show_id;
```

```
189  --Question 8:  Show every show and any viewing sessions on it. Shows that were never watched must still appear.
190  select A.show_id,
191  show_title,
192  session_id,
193  watch_minutes
194  from joins3.subs3.shows as A
195  left join joins4.subs4.sessions as B
196  on A.show_id = B.show_id;
197
```

Add parameter

Table +

	show_id	show_title	session_id	watch_minutes
1	701	Comedy Hour	904	20
2	702	Crime Time	903	60
3	703	Tech Tales	902	30
4	704	Cooking Lab	null	null
5	706	Wild Earth	null	null
6	701	Comedy Hour	901	45

Question 9

```
Select session_id,
show_id,
watch_minutes,
A.user_id,
user_name
From sessions AS A
Left Join users AS B
ON A.user_id = B.user_id;
```

```sql
--Question 9: Show every viewing session and the user who watched it. Sessions referencing users that do not exist must still appear (with NULL user details).
Select session_id,
show_id,
watch_minutes,
A.user_id,
user_name
From joins4.subs4.sessions AS A
Left Join joins.subs.users AS B
ON A.user_id = B.user_id;
```

Add parameter

Table

	session_id	show_id	watch_minutes	user_id	user_name
1	901	701	45	1	Nomvula
2	902	703	30	2	David
3	903	702	60	1	Nomvula
4	904	701	20	7	null
5	905	705	90	3	Anele

Question 10

select A.user_id,
user_name,
B.plan_id,
plan_name,
monthly_price
from users as A
left join subscriptions as B
on A.user_id = B.user_id
left join plans as C
on B.plan_id = C.plan_id;

```sql
--Question 10: Show every user, the plan they are on (if any), and the monthly price. Users without a subscription must still appear.
select A.user_id,
user_name,
B.plan_id,
plan_name,
monthly_price
from joins.subs.users as A
left join joins1.subs1.subscriptions as B
on A.user_id = B.user_id
left join joins2.subs2.plans as C
on B.plan_id = C.plan_id;
```

Add parameter

Table

	user_id	user_name	plan_id	plan_name	monthly_price
1	1	Nomvula	12	Premium	199
2	2	David	11	Standard	129
3	3	Anele	13	Family	249
4	4	Kabelo	null	null	null
5	5	Lerato	null	null	null
6	1	Nomvula	10	Basic	79

Part C

Question 11

```
select A.user_id,
user_name,
subscription_id,
start_date
from users as A
full join subscriptions as B
on A.user_id = B.user_id;
```

```
220
221 --Question 11:Show every user and every subscription, including users without subscriptions AND subscriptions referencing users that do not exist.
222 select A.user_id,
223 user_name,
224 subscription_id,
225 start_date
226 from joins.subs.users as A
227 full join joins1.subs1.subscriptions as B
228 on A.user_id = B.user_id;
229
```

Add parameter

Table

	user_id	user_name	subscription_id	start_date
1	5	Lerato	null	null
2	1	Nomvula	503	2026-03-10
3	3	Anele	505	2026-04-05
4	2	David	502	2026-02-01
5	4	Kabelo	null	null
6	1	Nomvula	501	2026-01-15
7	null	null	504	2026-03-20

Question 12

```
select A.plan_id,
plan_name,
subscription_id,
user_id
from plans as A
full join subscriptions as B
on A.plan_id = B.plan_id;
```

```
229
230 --Question 12: Show every plan and every subscription, including plans without subscribers AND any subscription referencing a plan that does not exist.
231 select A.plan_id,
232 plan_name,
233 subscription_id,
234 user_id
235 from joins2.subs2.plans as A
236 full join joins1.subs1.subscriptions as B
237 on A.plan_id = B.plan_id;
```

Add parameter

Table

	plan_id	plan_name	subscription_id	user_id
1	12	Premium	503	1
2	10	Basic	501	1
3	13	Family	505	3
4	14	Mobile	null	null
5	11	Standard	504	6
6	11	Standard	502	2

Question 13

```
select A.show_id,
show_title,
session_id,
watch_minutes
from joins3.subs3.shows as A
full join joins4.subs4.sessions as B
on A.show_id = B.show_id;
```

```
239
240  --Question 13: Show every show and every viewing session, including shows that were never watched AND sessions referencing shows that do not exist.
241  select A.show_id,
242  show_title,
243  session_id,
244  watch_minutes
245  from joins3.subs3.shows as A
246  full join joins4.subs4.sessions as B
247  on A.show_id = B.show_id;
```

Add parameter

Table +

	show_id	show_title	session_id	watch_minutes
1	704	Cooking Lab	null	null
2	703	Tech Tales	902	30
3	701	Comedy Hour	904	20
4	702	Crime Time	903	60
5	706	Wild Earth	null	null
6	701	Comedy Hour	901	45
7	null	null	905	90

Question 14

```
Select session_id,
show_id,
watch_minutes,
A.user_id,
user_name
From sessions AS A
Full Join users AS B
ON A.user_id = B.user_id;
```

```sql
--Question 14: Show every user and every viewing session, including users with no sessions AND sessions referencing users who do not exist.
Select session_id,
show_id,
watch_minutes,
A.user_id,
user_name
From joins4.subs4.sessions AS A
Full Join joins.subs.users AS B
ON A.user_id = B.user_id;
```

Add parameter

Table

	session_id	show_id	watch_minutes	user_id	user_name
1	901	701	45	1	Nomvula
2	903	702	60	1	Nomvula
3	905	705	90	3	Anele
4	902	703	30	2	David
5	904	701	20	7	null
6	null	null	null	null	Kabelo
7	null	null	null	null	Lerato

Question 15

select A.user_id,
user_name,
B.plan_id,
plan_name,
monthly_price,
subscription_id
from users as A
full join subscriptions as B
on A.user_id = B.user_id
full join plans as C
on B.plan_id = C.plan_id;

```sql
--Question 15: Show every user, every subscription, and every plan in one query — using FULL OUTER JOIN throughout. This is the hardest question — get all gaps visible at once.

select A.user_id,
user_name,
B.plan_id,
plan_name,
monthly_price,
subscription_id
from joins.subs.users as A
full join joins1.subs1.subscriptions as B
on A.user_id = B.user_id
full join joins2.subs2.plans as C
on B.plan_id = C.plan_id;
```

Add parameter

Table

	user_id	user_name	plan_id	plan_name	monthly_price	subscription_id
1	1	Nomvula	12	Premium	199	503
2	1	Nomvula	10	Basic	79	501
3	3	Anele	13	Family	249	505
4	5	Lerato	null	null	null	null
5	4	Kabelo	null	null	null	null
6	2	David	11	Standard	129	502
7	null	null	11	Standard	129	504
8	null	null	null	Mobile	59	null